

Rebuilding Serum 2: Technical Analysis and a Claude Code Implementation Spec for a Preset-Compatible JUCE Synth

You can build an original JUCE/CMake hybrid synth that sounds and works like Serum 2 and loads its presets. Serum 2's `.SerumPreset` container has been fully reverse-engineered: an `XferJson` header, a JSON metadata block, then zstd-compressed CBOR. [github +2](#) The Serum 1 `.fxp` container is well understood. [github](#) The weak point is that **parameter semantics** (value curves, enum orderings, some module internals) are not publicly documented, so your plan has to measure them against a licensed Serum 2. [github](#)

TL;DR

- **Feasible, with one hard part.** The Serum 2 container is solved: `XferJson\0` + u64 JSON length + JSON + u32 CBOR size + u32 version (=2) + zstd(CBOR). The CBOR is a map of about 160 modules (`Oscillator0..4`, `Env0`, `LFO*`, `ModSlot*`, `FXRack0..2`, `Macro0..7`...), each holding `plainParams`. [pypi](#) The hard part is mapping values (curves, enum indices, "default" sentinels) onto your engine, which needs a harness that hosts real Serum 2 and reads its parameter strings.
- **The scope is large but bounded.** It covers 3 main oscillators (Wavetable, Multisample/SFZ, Sample, Granular with up to 256 grains, Spectral), plus sub and noise. It also covers 2 filters (serial/parallel), 4 envelopes, up to 10 LFOs (including Lorenz/Rossler chaos), 8 macros, an arpeggiator and clip sequencer (12 slots per bank), and 13 effects plus 3 splitters across Main, Bus 1 and Bus 2. [xferrecords](#)
- **Build it in phases, clean-room, without GPL code.** Copying Vital or Surge code makes your plugin GPLv3. Use published algorithms (Zavalishin TPT/ZDF, Huovilainen ladder, Dattorro plate, mipmapped FFT wavetables) and permissive libraries (zstd, zlib, nlohmann::json CBOR, HIIR, xsimd). Never ship Xfer's names, artwork or factory content.

Assumptions (stated per request)

- Framework: JUCE 8 with CMake, C++20 (C++17-compatible core DSP).
- Primary target: VST3 on Windows x64 for FL Studio. Secondary: CLAP (via `clap-juce-extensions`) and AU.
- Preset compatibility: Serum 2 `.SerumPreset` and legacy Serum 1 `.fxp`/`.fxb`, matching the fact that Serum 2 itself imports Serum 1 presets. [monosounds](#) [github](#)
- Wavetables: must read and write Serum-style `.wav` wavetables with the `clm` chunk. [github](#)
- The product is original. It does not use the "Serum" or "Xfer" names, Xfer's UI artwork or layout, or any factory content.

Key Findings

1. **Serum 2's architecture is documented officially at feature level.** Xfer's "What's New in Serum 2" PDF (manual v1.0.0, dated March 17, 2025, for version 2.0) lists: [xferrecords](#)
 - three primary oscillators, each in Wavetable, Multisample, Sample, Granular or Spectral mode, plus a dedicated sub and noise oscillator; [xferrecords](#) [xferrecords](#)
 - dual filters with series or parallel routing and a new "Clean" drive mode; [sonic-weaponry](#) [xferrecords](#)
 - four envelopes with BPM sync and an invert-legato option; [xferrecords](#)
 - up to 10 LFOs ("LFO 7 to LFO 10 appear after you assign LFO 6"), with rates up to 1000 Hz; [xferrecords](#)
 - eight macros, each of which can also be a modulation destination; [xferrecords](#)
 - an arpeggiator and clip sequencer, each with 12 slots per bank; [xferrecords](#)
 - 13 effects plus 3 splitter modules across multiple buses. [xferrecords](#)
2. **The development credits hint at the DSP lineage, though this is inference.** The same PDF credits "Laurent de Soras" and "Andrew Simper" for special assistance. [xferrecords](#) De Soras wrote the HIIR polyphase half-band library; Simper (Cytomic) published widely used trapezoidal/ZDF SVF derivations. Dave Gamble is on the core team. In

KVR's "Official Serum 2 thread!" on March 18, 2025, he wrote: "MG Ladder and EMS Ladder do not have circuit modelled distortion. Wsp, Exp, Acid Ladder and MG Dirty do." Expect half-band polyphase oversampling and trapezoidal SVF/ladder topologies inside Serum 2, but treat this as informed inference.

3. **The `.SerumPreset` layout has been confirmed byte-for-byte by several tools, though they are not fully independent of each other.**
 - KennethWussmann's `serum-preset-packager` (Python) came first. Oxdevalias's gist says that as of 2025-04-22 Wussmann had "basically figured out completely" the v2 format.
 - CharlesBT's `node-serum2-preset-packager` (TypeScript) is a port based on that work.
 - `serum2-preset-loader` v0.1.3 on PyPI (MIT, tested against Serum 2.1.4) is a third implementation.
 - The processor-state variant adds a `hash` = `md5(compressed_cbor)` in its JSON header. Its CBOR differs from the preset CBOR in three known places. For example, the processor uses an empty map where presets use `"default"`.
4. **The Serum 1 `.fxp` layout is mostly mapped.** The MIT-licensed Rust tool `pycabbage/flp-extract-fxp`, validated against 25 real files and against static analysis of `Serum2.vst3` 2.0.23, documents: `github`
 - a 60-byte big-endian VST2 header (`CcK`/`FPCh`/`XfsX`); `github`
 - zlib stream 0 holding a 172,736-byte state, then further zlib streams for embedded wavetables, noise and filter tables, then a u32 LE trailer; `github`
 - preset name at 0x4972, format version (f32) at 0x4994, author at 0x49A0, category at 0x49D0; `github`
 - Serum 2's internal converter, `s1state_load`. `github` `github`
5. **The wavetable format is open and simple.** Steve Duda documented the `c1m` RIFF chunk (`<1>2048 01000000 wavetable (www.xferrecords.com)`). `xferrecords` `github` Frames are 2048 samples, with up to 256 frames. `github` `kvraudio` The second flag marks factory tables; Duda asks third parties to leave it at zero. `xferrecords`
6. **Exact menu lists and value curves remain the main unknowns.** The VST3 parameter dump shows a Filter Type default of normalized 0.009434 ($\approx 1/106$) for "MG Low 12". `github` If evenly spaced, that implies about 107 filter types, up from about 89 in Serum 1 (1/88). Filter Freq 0.5 displays "425" Hz; `github` an exponential curve would fit a range of about 8.2 Hz to 22.05 kHz ($\sqrt{8.19 \cdot 22050} \approx 425$). This is a hypothesis to verify.

Details — Part 1: Technical Analysis of Serum 2

1.1 Architecture and signal flow

Element	Serum 1 (manual v1.0.1)	Serum 2 (official PDF/web manual)
Main oscillators	2 wavetable (A, B)	3 (A, B, C); each Wavetable / Multisample / Sample / Granular / Spectral
Sub / Noise	Sub osc + stereo noise sample player	Both retained
Filters	1 per-voice + 1 FX filter	2 per-voice (series/parallel), plus FX filter
Envelopes	3 (Env1 hard-wired to amp)	4, BPM option, invert legato
LFOs	4 + 2 global Chaos	Up to 10; Lorenz and Rossler modes; up to 1000 Hz
Macros	4	8 (each also a destination)
Mod matrix	16 slots	VST3 dump lists "Mod 1-64", so 64 slots (inference)
FX	10 fixed modules, reorderable	13 effects + 3 splitters, multiple instances, Main + Bus 1 + Bus 2
Sequencing	none	Arp + Clip sequencer, 12 slots/bank, MIDI out
Formats	VST2/VST3/AU/AAX	VST3, AU, AAX 64-bit

Routing (official web manual). Every source (OSC A/B/C, SUB, NOISE) and both filters have a routing target:

- **Filter:** a balance knob from fully Filter 1 to fully Filter 2.
- **Main:** through the effects.
- **Direct:** bypasses filters and effects.
- **None:** silent; used as a modulation source only.

Each source also has send knobs to **BUS 1** and **BUS 2**. By default OSC A goes to Filter 1, which starts disabled; everything else goes to Main. The VST3 dump confirms per-source ">Filter Balance", ">BUS1" and ">BUS2" parameters, and global "Direct Vol", "Bus 1 Vol" and "Bus 2 Vol".

Voice architecture.

- Serum 1 counts each unison voice as a voice, up to 34 per note (2×16 unison + sub + noise). The manual calls polyphony of 8 "enough" and 16 "a bunch". ([amazonaws](#))
- Global controls: Mono, Legato, Porta Time, Porta Curve, Porta Always, Porta Scaled (VST3 IDs 3, 4, 10-13). ([github](#)) Scaled portamento uses the set time for a one-octave glide and scales with interval. ([amazonaws](#))
- The Global page "Voice control" section offsets pan, pitch and cutoff for up to 8 consecutive notes (EDMProd). ([edmprod](#))
- Serum 1 always applied "Limit Same Note Poly to 1"; Serum 2 makes it selectable, default off. Hercelot's "Deep Dive into Serum 2 (Part 4)" on note.com puts the default Poly Count at 8, raisable to 32.

MPE, tuning and quality.

- MPE: rekkerd's 2.0.17 notes include MPE Y fixes. The processor state includes ([tuningData](#)), ([github](#)) and later builds added an "option to apply tuning to pitch expression". ([rekkerd](#)) ([plugintorrent](#))
- **MTS-ESP support is not confirmed in sources I could verify.** Treat it as optional, via ODDSound's MIT-licensed MTS-ESP client.

- Forum posts mention "Good, High or Ultra" quality, an "S1 Compatibility mode" and "Disable Smoothing". In Serum 1, oversampling applied only to warps: [amazonaws](#) "Otherwise the standard oscillator quality is always maximum!" [amazonaws](#) [xferrecords](#)

1.2 Wavetable oscillator

Confirmed facts.

- Frames are 2048 samples, up to 256 frames (maximum $2048 \times 256 \times 32$ bits = 2 MB). [github](#) Morph tables are generated at load time; only the morph *type* is stored. [amazonaws](#)
- Unison: up to 16 voices with gain compensation; Detune and Blend (Blend default 75% = even mix). [amazonaws](#)
- Global unison: Range 0-48 semitones (default 2), Width, Warp offset, WT Pos offset, Stack (Off, 12 (1x), 12 (2x), +7 options, Center-12, Center-24), Tuning (Linear, Super, Exp, Inv, Random). [amazonaws](#)
- Serum 2 adds "Smooth Interpolation" (continuous frame position without morph tables), tuning modes (semitone, harmonics, ratio, step) and "Phase Memory". [sonic-weaponry +2](#)
- Serum 2 unison parameters: Unison, Uni Stack, Uni Detune, Uni Blend, Uni Width, **Uni Span**, **Uni Rand Start**, Uni Warp, **Uni Warp 2**. [github](#)
- Phase: 0-100%, where 100% shows "Mem" (continue from the previous note). Random is a per-voice random phase offset. [amazonaws](#)

Anti-aliasing (inference). Serum 1 claims no audible aliasing up to the limit of hearing, at extra load-time and run-time CPU cost, using SSE. [amazonaws](#) That fits **FFT-mipmapped band-limited tables**:

- per frame, take a 2048-point FFT; for each of about 11 octave levels, zero bins above the Nyquist limit for that level's highest pitch, then inverse-FFT;
- at run time, pick two adjacent levels from the phase increment and crossfade;
- read with cubic/4-point Hermite interpolation; Smooth Interpolation = bilinear across frames $\times$ within the sample.

Serum 1 warp modes (official list) and implementation hypotheses. $\boxed{p}$ is phase in $[0,1)$, $\boxed{k}$ is warp amount in $[0,1]$, $\boxed{w(p)}$ is a table read.

Mode	Official description (paraphrased)	Proposed math (inference)
Sync (Self Sync)	Plays faster, hard restarts at note period	$p' = \frac{p \cdot (1 + k \cdot (R-1))}{R}$, $R \approx 8-16$
Window Sync	Sync with a smoothing window	$y = W(p') \cdot \text{win}(p)$, win = Hann or $\sin(\pi p)$
Bend + / - / +/-	Pinch toward / away from the middle	piecewise power about 0.5: $p < .5 : .5 \cdot (2p)^e$, $e = 2^{(\pm c \cdot k)}$
PWM	Pushes the waveform left	$p' = p / (1 - k \cdot 0.99)$; output 0 when $p' \geq 1$
Asym + / - / +/-	Bend the whole cycle right / left	$p' = p^e$, $e = 2^{(\pm c \cdot k)}$
Flip	Polarity flip at a set position	$y = (p < k) ? W(p) : -W(p)$
Mirror	Second half mirrors the first (pressreader)	$p < .5 ? W(f(2p)) : W(f(2-2p))$, $f = \text{asym}(k)$
Remap 1-4	Drawable curve; mirrored; sinusoidal; 4×	$p' = \text{lerp}(p, C(p), k)$; C per half / $(0.5 - 0.5 \cos(\pi p)) / 4$ segments
Quantize	Sample-and-hold on the waveform	$p' = \text{floor}(p \cdot N) / N$, N from 2048 down to 2 (exponential in k)
FM (from other osc)	Serum 1 "FM" was actually phase modulation	$p' = \frac{p + k \cdot D \cdot m(t)}{D}$
AM / RM	Amplitude / ring modulation	$y \cdot (1 + k(m-1))$; $y \cdot \text{lerp}(1, m, k)$
FM Noise / FM Sub	PM from the noise / sub oscillator	as above, m = noise or sub

Serum 2 warp changes.

- Dual warp: two warp slots, with "Uni Warp" and "Uni Warp 2".
- "True FM": through-zero modulation of the phase increment.
- "PD from filters" and distortion warps.
- A third-party tutorial lists the warp menu headings as Off, Sync, Alt Warp, Filter, Distortion, FM, PD, AM, RM, Swap Warps. [pycabbage](#) shows the internal enum starts ([kNoWarp](#)), ([kSync](#)), ... ([github](#)) The exact list is **unverified** (§5.10).
- Spectral mode has its own "spectral warps", including Mask and Vocode from another oscillator (KVR review). ([kvraudio](#))

1.3 Wavetable editor features that affect compatibility

- **Morph types:** Crossfade; Spectral (magnitude + phase); Spectral (zero fundamental phase); Spectral (zero all phases). Morphing fills the frames between keyframes up to 256. ([amazonaws](#))
- **Import modes:**
 - normal (dynamic pitch zero-snap / follow);
 - constant frame size, by pitch average or by manual entry of 1-4 digits or a note name (assumes 44.1 kHz);
 - FFT 256/512/1024/2048;
 - a sidecar ([.txt](#)) override ([\[N\]](#) with 32-9999, optional ([\[no interp\]](#)));
 - Serum 2 adds **Frequency Estimation** for sample→wavetable conversion. ([xferrecords](#))
- **Formula parser:** generates single frames or whole tables. Documented variables include ([in](#)), ([q](#)) (frame index) and ([z](#)) (example ([z=\(q>1\)?in:0](#))), plus a ternary operator. Implement with a small Pratt parser or permissive ([exprtk](#)); verify any further functions.
- **Export:** mono 32-bit float, 44.1 kHz, 2048 samples per frame, with the ([c1m](#)) header. ([amazonaws](#))

`.wav` `c1m` **chunk (confirmed).**

- Layout: chunk ID `c1m` (0x63 0x6C 0x6D 0x20), a u32 LE size, then ASCII such as `<!--2048 01000000 wavetable (www.xferrecords.com)-->`.
- `2048` is the frame size. Duda said 0064-8192 would be supported later.
- Of the 8-digit flags, a community gist says the first digit is the interpolation/morph mode and the **second** is the factory flag. Write `0` there.
- Pad the chunk to even length. Vital reportedly needs this, and RIFF requires it.

1.4 New Serum 2 oscillator types

- Multisample:**
 - Loads **SFZ** ("open standard and human-readable"). Changelog items mention `loop_crossfade` and `default_path`.
 - Implement `<region>`, `<group>`, `<control>` `default_path`, `sample`, `lokey/hikey/key/pitch_keycenter`, `lovel/hivel`, `tune/transpose`, `loop_mode/loop_start/loop_end/loop_crossfade`, `amp_velcurve`, `seq_length/seq_position`, `trigger=release`.
- Sample:**
 - Modulatable start/end and loop points; loop crossfade.
 - Loop modes: VST3 default "One-shot", plus a forward-reverse tails mode; "snap loop detection".
 - Rate (tape-stop), Scan Rate / Scan BPM Rate / Scan Key Track / Position.
 - Auto and manual slicing (Single Slice; Slice Play Mode default "To Next Marker").
 - FM/PM/distortion warps.
 - DSP: 8-16-tap windowed-sinc with mipmaps by ratio; equal-power loop crossfade.
- Granular:**
 - Up to 256 grains; position, size, density/spacing, pitch, randomization; "Window Amount"; warps. `xferrecords`
 - DSP: asynchronous scheduler with a 256-grain pool (read position, rate, window, pan); table windows (Hann, Tukey(α =amount), Gaussian, triangle, rectangular).
- Spectral:**
 - "analyzing and resynthesizing the frequency spectrum... independent control over time and pitch".
 - Controls: Timbre Shifting, Scan Rate, Hi/Low Freqs, Envelope, `xferrecords` transient detection, drawable spectral filter, unison, standard plus spectral warps.
 - DSP: phase vocoder (2048-4096 FFT, 4× overlap) with phase locking; bin remapping for pitch; cepstral/LPC envelope warping for timbre; spectral-flux transient detection with phase reset.

1.5 Filters

- Serum 1 has roughly 90 modes** (PausePlayRepeat counts "ninety") in four categories, **Normal, Multi, Flanges, Misc**:
 - Normal: Low/High/Band/Peak/Notch 6-24 (state-variable) and MG Low 6-24 (ladder).
 - Multi: morphs (L/B/H, L/P/H, L/N/H, B/P/N) and duals (Var = second cutoff).
 - Flanges: comb/flanger and phaser variants.
 - Misc: Formant I/II/III, German LP (ZDF), French LP (two cascaded LPFs, "BOEUF" Var), Reverb (FDN), SampHold, ring mod and downsampling types.
- Controls:** Cutoff, Res, Stereo (L/R cutoff offset), Drive, Var, Mix. Keytrack is one octave of filter per MIDI octave. Serum 2 adds X/Y (PZ SVF morphing), Level `qithub` and the "Clean" drive mode. `xferrecords` `edmprod`
- New Serum 2 types:** MG Ladder, Acid Ladder, EMS Ladder, Wsp, Exp, MG Dirty, PZ SVF (draw and morph several responses), Diffuser/Disperser (allpass chain), Comb 2, and a DJ-mixer-style filter. `cfa-sound +4`

- Dave Gamble: "MG Ladder and EMS Ladder do not have circuit modelled distortion. Wsp, Exp, Acid Ladder and MG Dirty do." ([kvraudio](#))
- Monosounds, a Serum pack designer since 2022, counts "Eleven new types, including analog-style emulations that actually saturate like hardware."

Family	Algorithm	Reference
SVF LP/HP/BP/Notch/Peak/All	TPT/trapezoidal SVF, $g = \tan(\pi fc/fs)$, $k = 2-2r$	Zavalishin, <i>The Art of VA Filter Design</i> (rev. 2.1.2); Simper/Cytomic SVF papers
6/18 dB slopes	1-pole TPT, cascaded with a 2-pole	Zavalishin ch. 3
Ladder (MG Low, MG Ladder)	$4 \times$ 1-pole TPT with zero-delay global feedback	Huovilainen (DAFx 2004); Välimäki & Huovilainen (CMJ 2006); Stilson & Smith (ICMC 1996)
Driven ladders (Acid, MG Dirty)	Per-stage tanh/OTA nonlinearity, Newton-Raphson, $2 \times$ oversampled	Huovilainen; D'Angelo & Välimäki
Acid (303-style diode ladder)	Diode-ladder ZDF with mismatched stage caps	Zavalishin ch. 5; Stinchcombe's TB-303 analysis
Wsp / EMS	Nonlinear TPT SVF with inverter-style saturation / alternate ladder coupling	inference
MS-20/Sallen-Key	Nonlinear Sallen-Key ZDF	Zavalishin §5; Pirkle
Comb/Flange	Fractional delay (allpass/Lagrange), feedback $\pm$, delay = $1/fc$	Smith, <i>Physical Audio Signal Processing</i>
Phaser filters	N first-order TPT allpasses + feedback	Zavalishin
Formant/Vowel	3-5 parallel BP SVFs over vowel formant tables	Peterson & Barney data
Reverb filter	Small FDN (4-8 lines, Householder) tuned by cutoff	Jot & Chaigne 1991
SampHold/Downsample	ZOH decimation	—
Disperser/Diffuser	8-64 cascaded 2nd-order allpasses at cutoff	—
PZ SVF	User poles/zeros $\rightarrow$ biquad cascade; morph pole/zero positions	inference

1.6 Modulation system

- **Envelopes:** Attack, Hold, Decay, Sustain, Release, plus Atk/Dec/Rel curves; envelopes 2-4 add Start/End. ([github](#)) Env1 drives amplitude. ([edmprod](#)) The default preset shows Attack ≈ 0.000515 ([pypi](#)) and Decay ≈ 0.966 in ([plainParams](#)), likely seconds (inference). ([pypi](#))
- **LFOs:**
 - Drawable breakpoint shapes, independent X/Y grid, shape presets.
 - Modes: Trig, Env (one-shot with loopback), Off (free-running).
 - BPM sync with dot/triplet and Anchor; Rate, Rise, Delay, Smooth. Serum 2 adds Phase, swing-follow and directional playback.

- Lorenz/Rossler appear in `LFO{k}.plainParams.kParamType`. [github](#) Serum 1 Chaos 1/2 have BPM sync and a Mono switch. [amazonaws](#)
- **Sources:** velocity, note, aftertouch, mod wheel, pitch bend, Note-On Random 1/2, Fixed, noise osc output, MPE. In Serum 2 "any oscillator or filter can be a modulation source" (audio-rate).
- **Matrix slot:** Source, bipolar Amount, Destination, Type (uni → / bi ↔), Aux Source (multiplies; "(inv)" variant), Aux Amount, Output scale, Bypass.
 - Serum 2 adds editable source and aux curves, reordering and removal.
 - Drag-and-drop onto a centered knob is bidirectional, otherwise unidirectional.
- **Per-voice vs. global:** FX run on the summed signal, so per-voice sources retrigger FX modulation on each note. [amazonaws](#) LFOs can be per-voice or global (implied by the 2.0.17 "non-per-voice param" fix). [rekkerd](#)

1.7 Arpeggiator and clip sequencer

- **Arp:** pattern/shape (with a "Pattern" editor), transpose shift and range, offset, repeats, gate, chance, retrigger modes, velocity ramp, [github](#) MIDI Out; 12 slots per bank (`.XferArpBank`). [github](#) [xferrecords](#)
- **Clip:**
 - Piano roll, flexible grid, Overdub/Extend recording, loop and offset markers, automation lanes, MIDI Out, 12 slots per bank.
 - Key, scale, transpose and swing apply to MIDI input and clip/arp output.
 - An osc mapping editor sets key and velocity ranges per oscillator and arp.
- **CBOR modules:** `Arp0`, `MidiClip0..11`, `ClipPlayer`, `PitchQuantizer{n}` (with `scaleName`). [pypi](#)

1.8 Effects rack

- **Structure:** 13 effect types and 3 splitters (L/H, L/M/H, M/S) across `FXRack0..2` (Main, Bus 1, Bus 2). Instances are unlimited and reorderable, with rack and module presets. The VST3 dump exposes 16 generic "FX Main / Bus 1 / Bus 2 Params". [github](#)

FX	Known parameters/modes	Reproduction DSP
Distortion	13 Serum 1 modes incl. Tube, Downsample, X-Shaper, X-Shaper (Asym); Serum 2 adds Overdrive + DC Bias; pre/post filter, LP→BP→HP morph, F, Q, Drive	Waveshapers at 4× (HIIR); X-Shaper = crossfade of two drawn curves; Downsample = ZOH
Flanger	BPM sync, Rate 0–20 Hz, Depth, Feed, Phase	Modulated fractional delay 0.1–10 ms
Phaser	Rate, Depth, Freq, Feed, Phase	4–12 TPT allpass stages
Chorus	4 voices (2 L, 2 R), Delay 1/2, Depth, Feed, LPF	Modulated delay lines
Delay	Normal / Ping-Pong / Tap→Delay; BPM, Link, L/R time + offset (TRIP 1.333, DOT 1.5), BP filter; HQ mode default	Fractional delay; BP SVF in feedback
Compressor	Thresh 0 to –120 dB, Ratio (max = true-peak "Limit"), Attack, Release, Gain (~36 dB), Multiband up/down	Feed-forward detector; multiband = 3-band Linkwitz-Riley up/down ("OTT-style")
Reverb	Plate (modified TAL-Reverb) in Serum 1; Serum 2 adds Vintage , Nitrous , Basin ; Size, Pre Delay, Low Cut, Damp, Hi Cut, Width	Dattorro plate (JAES 1997); FDN variants, tuned by ear
EQ	2 bands; low: shelf/peak/HP; high: shelf/peak/LP	RBJ biquads or TPT shelves
Filter	Any filter type as master effect	Reuse filter module
Hyper/Dimension	Hyper: 1–7 voice micro-delay chorus, Rate, Detune, Retrig, Unison, Mix; Dimension: 4 delay lines summed out of phase with slow AM, Size, Mix	As documented
Bode	Frequency shifter	Hilbert (HIIR polyphase pair) × complex oscillator
Convolve	IR loading (users embed IRs in presets)	Uniform-partitioned FFT convolution
Utility	Phase invert, bass mono, stereo imaging	Trivial
Splitters	L/H, L/M/H crossovers; M/S	LR4 crossovers

1.9 Performance engineering

- Process voices in **SIMD lanes** (4 or 8 unison streams per `xsimd::batch<float>`), SSE2/AVX2), with structure-of-arrays oscillator state.
- Compute modulation per 16–32-sample block with linear ramps; run audio-rate sources per sample only on routed destinations.
- Smooth every host parameter (5–20 ms), with a "disable smoothing" option. Use `juce::ScopedNoDenormals` and DC-safe feedback.
- Oversample (HIIR 2×/4×) only around nonlinear stages; map Good=1×, High=2×, Ultra=4× (inference).
- No allocation on the audio thread. Preallocate voice/grain pools and FFT buffers; load assets on a background thread and publish them via atomic `shared_ptr` swap.

Details — Part 2: Preset and File Formats

2.1 `.SerumPreset` (confirmed byte layout)

offset	size	field
0	9	magic "XferJson\0" (58 66 65 72 4A 73 6F 6E 00)
9	8	u64 LE N = JSON metadata length
17	N	UTF-8 JSON e.g. {"fileType":"SerumPreset","presetName":...,"presetAuthor":...,"tags":[...],...}
17+N	4	u32 LE uncompressed CBOR size
21+N	4	u32 LE container format version (=2)
25+N	M	Zstandard frame containing CBOR (RFC 8949)

- **Processor state** uses the same container, with JSON
`{"component":"processor","product":"Serum2","productVersion":"2.1.4","hash":md5(compressed CBOR),...}`. `(pypi)`
- **CBOR root:** about 160 module keys (`(Arp0)`, `(Env0)`, `(Oscillator0..4)`, `(Macro0..7)`, `(ModSlot*)`, `(LFO*)`, `(FXRack0..2)`, `(MidiClip0..11)`, `(RoutingSlot*)`, `(PitchQuantizer*)`).
 - Each module holds `(plainParams)`: either the string `("default")` or a map of `(kParam*)` keys to numbers.
 - Modules also carry extras such as `(curveData)`, `(pathData)`, `(clip)`, `(embeddedNoiseData)`, `(pathToNoiseSample)` and `(storedPhasePos)`.
- **Indices:**
 - `(Oscillator0..2)` = A/B/C, `(Oscillator3)` = Noise (`(Noise0osc3)`), `(Oscillator4)` = Sub (`(Sub0osc4)`).
 - `(ModSlot{k})` uses `(destModuleID)`/`(destModuleParamID)` (-1 = empty); envelope sources are `([38,0])`.
 - `(RoutingSlot*.kParamRoutingDest)` $\in$ `{(kRoutingDestMaster), (kRoutingDestDirect), ...}`.
- **Preset-only keys:** `(fileType)`, `(presetName)`, `(presetAuthor)`, `(presetDescription)`, `(arpBankDisplayName)`, `(clipBankDisplayName)`, `(pypi)` UI state (`(ClipPlayer)`, `(Filter)`, `(SerumGUI)`), and template lists (`(GranularOsc)`, `(MultiSampleOsc)`, `(Osc)`, `(SpectralOsc)`, `(WTOsc)`).
- **Processor-only keys:** `(component)`, `(killEnvsGracefullyCompat)`, `(Arp0.activeClip)`, `(productVersion)`, and `(version)` = 10.0 (CBOR float).
- **Embedding:** unedited factory content is referenced by path. Expect user wavetables, samples and IRs inline as CBOR byte strings; the exact keys are an open question.

2.2 Serum 1 `(.fxp)`/`(.fxb)`

```
0x00 'CcnK' 0x04 u32 BE byteSize (Serum writes full file length) 0x08 'FPCh'
0x0C u32 BE 1 (version) 0x10 'XfsX' (plugin ID) 0x14 u32 BE fxVersion=1
0x18 u32 BE numParams=1 0x1C char[28] prgName 0x38 u32 BE chunkSize = len-60
0x3C chunk: [zlib#0 → 172,736-byte state][zlib#1..n: wavetables/noise/filter tables][u32 LE = compressed
size of zlib#0]
```

- **State fields:** name @0x4972 (32 B), f32 version @0x4994 (e.g. 0.1631), author @0x49A0 (48 B), category @0x49D0 (48 B). `(github)`
- **Older states:** 21,808 B (v0.1470) and 28,232 B (v0.1531). Zero-pad them to 172,736 bytes as Serum 2 does. Their extra streams are ordered `([frames][tuning][noise])`.
- **Parameter array:** a community repo reports about 300 LE floats at state offset 0x3460 in `(SYParameters.txt)` order (e.g. 44 Fil Type, 45 Cutoff, 46 Reso, 47 Drive). `(github)` `(github)` It is a low-reputation source; verify with your own diffs.
- **Version-gated migrations:**
 - filter-type requantization for versions below 0.155 (K=88 below v0.142, else 89);

- mod-destination shifts below v0.0299;
- `mixOrGain` from v0.159;
- MPE above v0.155;
- Downsample re-indexing below v0.162.
- `.fxb`: a bank chunk (`FBCh`) or (`FxBk`) holding N (`FPCh`) programs; parse each as above.

2.3 Value mapping, and measuring the unknowns

- `plainParams` probably store plain (real-unit) values, while VST3 automation is normalized 0–1. Confirm both directions for each parameter.
- **Best method (no audio needed):**
 1. Host Serum 2 in a minimal VST3 harness (JUCE, DawDreamer or pedalboard).
 2. Sweep each parameter's normalized value 0...1 in 1/1024 steps and record `getParamStringByValue` (JUCE `getText`).
 3. Fit linear, exponential ($a \cdot (b/a)^x$), power ($a + (b-a)x^n$) or enum tables.
 4. Save presets with known values and diff the CBOR to link `kParam*` keys to VST3 IDs.
- **Audio verification:**
 - Render fixed-seed probes with unison random phase off.
 - Fit the –3 dB cutoff, envelope crossings and LFO periods.
 - Null-test: residual below –40 dB for linear paths; spectral distance for nonlinear paths.

Details — Part 3: Reference Implementations and Libraries

Project	License	Relevant parts (study, don't copy, unless you accept GPL)
Vital (mtytel/vital)	GPLv3 (commercial license via licensing@vital.audio) (github)	<code>src/synthesis/producers/synth_oscillator.*</code> (wavetable, unison, spectral morph), <code>wavetable/</code> , <code>filters/</code> , <code>effects/</code>
Surge XT	GPLv3	<code>src/common/dsp/oscillators/</code> , <code>sst-filters</code> , <code>sst-effects</code> (check sub-library licenses)
Dexed	GPLv3	FM operator, DX7 envelopes
Odin 2	GPLv3	Ladder/diode/SEM-style filters, wavetable osc
Helm	GPLv3	Simple voice manager, mod matrix

- **Licensing:**
 - Copying or linking GPLv3 code makes your plugin GPLv3. Vital's README also forbids use of its trademarks and iOS App Store distribution.
 - JUCE 8 is AGPLv3 or commercial. The JUCE 8 EULA tiers are Starter (free, up to \$20,000 annual revenue or funding), Indie (up to \$300,000; \$40/month or \$800 perpetual) and Pro (no limit; \$175/month or \$3,500).
 - Steinberg's VST SDK 3.8.0 changelog (2025/10/20) says "VST 3 is now open source, released under the MIT license."
 - CLAP is MIT.
- **Permissive libraries:** `zstd` (BSD-3/GPLv2), `zlib`, `nlohmann::json` (MIT; `from_cbor`/`to_cbor`) or `jsoncons`, `md5`, `HIIR` (WTFPL), `xsimd` (BSD-3), `pffft`/KissFFT or `juce::dsp::FFT`, `chowdsp_utils` (mostly BSD-3), `r8brain` (MIT), `Catch2`, `ODDSound` MTS-ESP client (MIT).
- **Books and papers:**

- Zavalishin, *The Art of VA Filter Design*.
- Pirkle, *Designing Software Synthesizer Plugins in C++*.
- Välimäki & Huovilainen (CMJ 2006).
- Dattorro, "Effect Design Part 1" (JAES 1997).
- Jot & Chaigne (1991).
- Smith, *Physical Audio Signal Processing and Spectral Audio Signal Processing*.
- Laroche & Dolson (1999).
- Roads, *Microsound*.

Details — Part 4: Legal/IP Notes (not legal advice)

- **Precedent:**
 - US courts have treated reverse engineering for interoperability as fair use (*Sega v. Accolade*, 1992; *Sony v. Connectix*, 2000).
 - EU Software Directive 2009/24/EC Art. 6 permits decompilation for interoperability. The CJEU in *SAS Institute v. World Programming* (2012) held that functionality and data file formats are not protected by copyright.
 - Vital and many Serum-like synths are distributed openly. `(alternativeto)` `(kvraudio)`
- **Risks:** EULA anti-reverse-engineering clauses (contract law), trademark misuse, UI trade-dress copying, factory-content copying.
- **Clean-room rules:**
 1. Never decompile Serum yourself; use published third-party format facts and black-box measurement.
 2. Use no code derived from Xfer binaries.
 3. Keep "Serum"/"Xfer" out of product names; only nominative "imports Serum® presets" with a disclaimer.
 4. Use an original UI and artwork.
 5. Never bundle Xfer factory wavetables, presets, samples, IRs or multisamples. Point users at their own licensed folder (`(Documents\Xfer\Serum 2 Presets\)`).
 6. Keep a dated log of where each fact came from.

Part 5 — DELIVERABLE: `(SPEC.md)` for Claude Code

Copy everything from here to the end of this section into `(SPEC.md)` at the repo root. Claude Code should treat it as the source of truth. Product codename: "**Tincture**"; rename freely.

5.0 Ground rules for the agent

- C++20, JUCE 8, CMake ≥ 3.25, MSVC 2022 on Windows x64 primary; clang on macOS secondary.
- Audio thread: no allocation, locks, file I/O or logging.
- Do not copy GPL code (Vital, Surge, Odin, Dexed, Helm); implement from the equations here.
- Every DSP module ships with a unit test and a golden render.
- Keep compatibility constants in `(src/compat/serum_tables.hpp)`, each marked `(// VERIFIED)`, `(// INFERRED)` or `(// TODO-MEASURE)`.

5.1 Repository layout

```
tincture/  
  CMakeLists.txt  cmake/  libs/ (JUICE, clap-juce-extensions, zstd, nlohmann_json, hiir, xsimd, pffft, Catch2)  
  src/  
    plugin/ PluginProcessor PluginEditor Parameters  
    engine/ Engine VoiceManager Voice SimdTypes.h Smoother.h  
    osc/ WavetableOsc WavetableBank Warp Unison.h SubOsc NoiseOsc SampleOsc MultisampleOsc+Sfz  
  GranularOsc SpectralOsc  
    filter/ Svf.h OnePole.h Ladder.h DiodeLadder.h SallenKey.h Comb.h Formant.h FdnFilter.h Disperser.h  
  PzSvf.h FilterBank (~107 entries)  
    mod/ Envelope.h Lfo.h Chaos.h ModMatrix Macros.h  
    seq/ Arp ClipPlayer PitchQuantizer.h  
    fx/ FxRack Distortion Flanger Phaser Chorus Delay Compressor Multiband Reverb(Plate,Fdn) Eq FilterFx  
  HyperDimension Bode Convolve Utility Splitters  
    io/ WavetableWav (clm) SerumPresetReader FxpReader Cbor.h Md5.h AssetResolver  
    compat/ serum_tables.hpp Serum2Mapper Serum1Mapper MigrationS1toS2  
    ui/ (original design)  
  tests/ unit/ golden/ presets/ (user-supplied, gitignored)  
  tools/ measure_host/ preset_dump/
```

5.2 CMake skeleton

```
cmake  
  
cmake_minimum_required(VERSION 3.25)  
project(Tincture VERSION 0.1.0 LANGUAGES C CXX)  
set(CMAKE_CXX_STANDARD 20)  
include(FetchContent)  
FetchContent_Declare(JUCE GIT_REPOSITORY https://github.com/juce-framework/JUCE.git GIT_TAG 8.0.4)  
FetchContent_Declare(zstd GIT_REPOSITORY https://github.com/facebook/zstd.git GIT_TAG v1.5.6 SOURCE_SUBDIR  
FetchContent_Declare(json GIT_REPOSITORY https://github.com/nlohmann/json.git GIT_TAG v3.11.3)  
FetchContent_Declare(xsimd GIT_REPOSITORY https://github.com/xtensor-stack/xsimd.git GIT_TAG 13.0.0)  
FetchContent_Declare(Catch2 GIT_REPOSITORY https://github.com/catchorg/Catch2.git GIT_TAG v3.6.0)  
FetchContent_Declare(clap-juce-extensions GIT_REPOSITORY https://github.com/free-audio/clap-juce-extensions)  
set(ZSTD_BUILD_PROGRAMS OFF) set(ZSTD_BUILD_SHARED OFF)  
FetchContent_MakeAvailable(JUCE zstd json xsimd Catch2 clap-juce-extensions)  
juce_add_plugin(Tincture  
  COMPANY_NAME "YourCo" PLUGIN_MANUFACTURER_CODE Ycoo PLUGIN_CODE Tnct  
  FORMATS VST3 AU Standalone IS_SYNTH TRUE NEEDS_MIDI_INPUT TRUE NEEDS_MIDI_OUTPUT TRUE  
  VST3_CATEGORIES Instrument Synth COPY_PLUGIN_AFTER_BUILD FALSE)  
target_sources(Tincture PRIVATE ${TINCTURE_SOURCES})  
target_compile_definitions(Tincture PUBLIC JUCE_VST3_CAN_REPLACE_VST2=0 JUCE_WEB_BROWSER=0 JUCE_USE_CURL=0)  
target_link_libraries(Tincture PRIVATE juce::juce_audio_utils juce::juce_dsp libzstd_static  
  nlohmann_json::nlohmann_json xsimd juce::juce_recommended_config_flags juce::juce_recommended_lto_flags)  
clap_juce_extensions_plugin(TARGET Tincture CLAP_ID "com.yourco.tincture" CLAP_FEATURES instrument synthesizer)
```

- zlib: JUCE's bundled `juce::GZIPDecompressorInputStream` or system zlib. Vendor HiIR and pffft into `libs/`.
- Windows:
 1. `cmake -B build -G "Visual Studio 17 2022" -A x64 && cmake --build build --config Release`

2. Copy `Tincture.vst3` to `C:\Program Files\Common Files\VST3\`.
 3. In FL Studio, Options → Manage plugins → Find more plugins.
- FL Studio: support variable block sizes (including 1 sample), handle `setStateInformation` before `prepareToPlay`, and report latency 0 unless Convolve or a lookahead limiter is active.

5.3 Architecture

- **Parameters:**
 - APVTS for host-automatable parameters only (about 1,500 at most), with stable string IDs (`oscA.level`).
 - Mirror Serum 2's VST3 ID blocks: osc n at 1000000+1000·n, filters 2000000+, envs 3000000+, LFOs 4000000+, mod slots 6000000+, macros 7000000+, routing 9000000+, FX slots 17000000+.
 - Non-automatable state (curves, wavetables, SFZ, clips, matrix topology) lives in a `PatchModel` tree.
- **Threading:**
 - The message thread edits `PatchModel` and publishes an immutable `EngineSnapshot` via atomic pointer.
 - The audio thread adopts the snapshot at block start. Old snapshots retire through a lock-free FIFO to a GC thread.
 - Asset loading and mipmap/FFT builds run on a `juce::ThreadPool`.
- **Voice manager:**
 - 64-voice pool, each voice up to 3×16 unison + sub + noise; polyphony 1-64.
 - Steal released/oldest voices first, then the quietest.
 - Mono/legato note stack (last/low/high); log-frequency portamento (time, curve, always, scaled); "limit same note".
 - MPE via `juce::MPEInstrument`.
- **Modulation:**
 - Per block, evaluate per-voice and global sources.
 - Each slot computes `out = curve_src(src) · amount · (aux ? curve_aux(aux)·auxAmt : 1)`. Bipolar maps `[0,1]→[-1,1]`.
 - Accumulate in normalized space, clamp, then apply output scale. Audio-rate slots run per sample for flagged destinations.
- **SIMD:** unison voices in 8-wide AVX2 lanes (SSE2 4-wide fallback) via `xsimd::batch<float>`; filters per voice with stereo packed.

5.4 Milestones and acceptance criteria

Phase	Scope	Acceptance criteria
1 MVP	1 WT osc, mipmapped tables, <code>c1m</code> wav import/export, SVF LP12, Env1 amp, poly 16	Loads any 2048×N wav; aliasing < -90 dBFS at C8 saw @48 kHz; pluginval strictness 10; loads in FL Studio
2 Oscillators	3 WT osc, sub, noise, unison (16, all stack/tuning modes, width, span, blend), all Serum 1 warps + dual warp, smooth interpolation, phase/random/mem, Good/High/Ultra	Unit tests per warp; 16 voices × 3 osc × 7 unison < 25% of one core @48 kHz/128
3 Modulation	4 envs, 10 LFOs, chaos, 8 macros, 64-slot matrix with aux and curves	Env timings ±1 sample; LFO at 1000 Hz alias-free
4 Filters	Full registry (~107 types), dual routing, drive/clean, var/X/Y, keytrack, stereo	Linear responses within 0.5 dB of analytic; stable self-oscillation; no NaN under fuzzing
5 FX	13 effects + 3 splitters, Main/Bus1/Bus2, reorder, multiple instances	Null tests for linear FX; reverb RT60 within 10%
6 Presets	<code>.SerumPreset</code> reader, <code>.fxp/.fxb</code> reader, mapping layer, asset resolver, own format	100% of corpus parses; ≥90% of mapped params within tolerance; A/B on 50 presets
7 New osc + seq	Sample, SFZ Multisample, Granular (256 grains), Spectral, Arp, Clip, key/scale, osc mapping	SFZ suite plays; granular CPU bounded; spectral ±12 st with < 1% formant drift
8 UI/polish	Original resizable UI, preset browser, undo/redo	No audio-thread allocations; 0 pluginval warnings; FL state recall

5.5 Module specifications

WavetableBank.

- Input: N frames (1-256) × F samples (default 2048). Accept 64-8192 and resample to 2048.
- Build: per frame, FFT, then for levels $L = 0..10$ keep harmonics $h \leq \text{floor}(1024 / 2^L)$ and inverse-FFT. Store `float[levels][frames][2048+4]` with wrap guards.
- Morph fill: crossfade (linear), spectral (linear magnitude, shortest-arc phase), zero-fund-phase, zero-all-phases.
- Level select: `L = clamp(floor(log2(max(1, inc·2048))), 0, 10)`, crossfading with L+1 by the fractional part.

WavetableOsc parameters

Name	Range	Default	Unit/curve	Notes
level	0-1	0.75	linear amplitude	TODO-MEASURE dB law
pan	-1..1	0	linear	constant-power
octave / semi / fine	-4..+4 / -12..+12 / ± 100	0	int / int / cents	
coarse	-48..+48	0	st (continuous)	
ratio / hzOffset	—	1 / 0	Serum 2 tuning modes	TODO-MEASURE
wtPos	0-1	0	$\rightarrow \text{frame} \cdot (N-1)$	smooth vs stepped
unison	1-16	1	int	
uniDetune	0-1	0.25	$\times \text{range}$ (0-48 st, default 2)	curve per tuning mode
uniBlend / uniWidth	0-1	0.75 / 1	linear	0.75 = even
uniSpan / uniRandStart	0-1	0	—	TODO-MEASURE
uniStack	Off, 12 $\times$ 1, 12 $\times$ 2, +7 variants, C-12, C-24	Off	enum	
warp1/warp2 mode, amount	enum, 0-1	Off, 0	—	§1.2
phase / random	0-1	0 / 1.0	linear	1.0 phase = Mem
pitchTrack	bool	on		

- Detune offsets for $t = 2i/(U-1) - 1$ (exponents INFERRED):
 - Linear: t .
 - Super: $\text{sign}(t) \cdot |t|^{\wedge 1.3}$.
 - Exp: $\text{sign}(t) \cdot |t|^{\wedge 2}$.
 - Inv: $\text{sign}(t) \cdot |t|^{\wedge 0.5}$.
 - Random: redrawn at note-on.
 - Semitones = offset·detune·range.
- Gain: $1/\sqrt{t(U)}$ with blend weighting; at blend = 0.75 all voices are equal.

Envelope.

- Stages: delay $\rightarrow$ attack $\rightarrow$ hold $\rightarrow$ decay $\rightarrow$ sustain $\rightarrow$ release.
- Curve: $y = (\exp(c \cdot x) - 1) / (\exp(c) - 1)$, $c \in [-10, 10]$.
- Times: 0-32 s, hypothesized $t = 32 \cdot x^{\wedge 3}$ (TODO-MEASURE).
- Options: BPM mode, legato invert, Start/End (envs 2-4).

LFO.

- Breakpoints (x, y, curve) with loopback; modes Trig/Env/Off.
- Rate: Hz 0.01-100 ($\times 10$ to 1000 Hz), or BPM 8 bars-1/64 with dot/triplet.
- Rise, delay, smooth, phase, anchor, swing, direction.
- Chaos: Lorenz ($\sigma=10$, $\rho=28$, $\beta=8/3$) and Rossler ($a=0.2$, $b=0.2$, $c=5.7$), integrated with RK4 and normalized by running min/max.

ModMatrix slot. `{src, srcCurve, amount -1..1, bipolar, dest, aux, auxCurve, auxAmount, auxInvert, outScale, bypass}`, 64 slots.

Filter module.

- Parameters: type (registry index), cutoff (8.18–22050 Hz exponential, INFERRED), res, drive (+clean), var/fat, stereo, X/Y, mix, level, keytrack.
- Registry entries: `{name, family, slopes, varMeaning, nonlinear, oversample}`.

FX, Sample/Granular/Spectral, Arp/Clip. Follow §1.4, §1.7 and §1.8.

- Oscillator block IDs: Start 10, End 11, Reverse 12, Scan Rate 13, Scan BPM Rate 14, Scan Key Track 15, Position 16, Loop Start 17, Loop End 18, Loop X-Fade 19, Loop Mode 20, Relative Loop 21, Single Slice 22, Slice Play Mode 23. ([github](#))
- Arp: up/down/updown/random/order/chord/Pattern, rate, gate, octave range, repeats, offset, chance, velocity ramp, retrigger, swing.
- Clip: 12 PPQ note slots, markers, automation lanes, MIDI out.

5.6 Preset compatibility spec

SerumPresetReader

```
cpp
Result<PatchModel> readSerumPreset(std::span<const uint8_t> b) {
    require(b.size() >= 25 && memcmp(b.data(), "XferJson\0", 9) == 0);
    uint64_t n = le64(b, 9); // JSON length
    auto meta = json::parse(b.subspan(17, n)); // fileType == "SerumPreset" or component == "processor"
    size_t o = 17 + n;
    uint32_t rawSize = le32(b, o), ver = le32(b, o + 4); // ver == 2 expected; warn otherwise
    auto comp = b.subspan(o + 8);
    std::vector<uint8_t> cbor(rawSize);
    require(ZSTD_decompress(cbor.data(), rawSize, comp.data(), comp.size()) == rawSize);
    json root = json::from_cbor(cbor, true, true, json::cbor_tag_handler_t::store); // keep byte strings
    return Serum2Mapper::map(root, meta);
}
```

- `plainParams == "default"` → apply the module's default table.
- Preserve unknown keys in `PatchModel.unknown`.
- Write `.SerumPreset` only in tests. When writing processor-style state, recompute the CBOR size and `hash = md5(zstd bytes)`.

FxpReader.

- Validate `CcnK`/`FPCh`/`XfsX`.
- Inflate zlib stream 0 until `Z_STREAM_END`, then loop through the following streams to the trailer.
- Zero-pad the state to 172,736 bytes.
- Read the version at 0x4994 and run `MigrationS1toS2` (§2.2).
- Map the parameter array (0x3460, LE f32, `SYParameters` order, INFERRED) via the Serum 1 names: `a_vol`, `a_pan`, `a_octave`, `a_semi`, `a_fine`, `a_unison`, `a_unidet`, `a_uniblend`, `a_warp`, `a_wtpos`, `b_*`, `noise_*`, `sub_osc_*`, `env1_*`, `fil_*`, `lfo1rate`..., `porttime`, `chaos1_*`, FX `verb_*`, `eq_*`, `dist_*`, `flg_*`, `phs_*`, `cho_*`, `dly_*`, `cmp_*`, `fx_fil_*`, `hyp_*`, `mod_N_amt/out`.
- Embedded wavetable streams inflate to float32 frames plus a morph header (one observed file has a 136-byte header; TODO-MEASURE).

Mapping table skeleton (complete with the harness)

Serum 2 module.key	S1 name	VST3 ID	Engine param	Transform	Status
Oscillator0 level (key TBC)	a_vol	1000001	oscA.level	plain→linear	TODO-MEASURE
Oscillator0 pan	a_pan	1000002	oscA.pan	±1	TODO
Oscillator0 octave/semi/fine	a_octave/a_semi/a_fine	1000003-5	oscA.pitch	int/int/cents	TODO
Oscillator0 unison/detune/blend	a_unison/a_unidet/a_uniblend	1000024/26/27	oscA.uni*	int/curve/lin	TODO
Filter0 type	fil_type	2000002	filter1.type	enum (S1: 89, K=88/89 requant; S2 ≈107)	PARTIAL
Filter0 freq	fil_cutoff	2000003	filter1.cutoff	Hz = 8.18·(22050/8.18)^x	VERIFY
Filter0 res/drive/var/stereo	fil_reso/driv/var/stereo	2000004-7	filter1.*	%	TODO
Env0.kParamAttack/Decay/...	env1_atk/dec/...	3000000-4	env1.*	seconds	LIKELY
LFO{k}.kParamType	—	—	lfoK.type	enum incl. Lorenz/Rossler	PARTIAL
ModSlot{k}.destModuleID/ParamID	mod_N_out	6000000+	matrix[k].dest	ID lookup	TODO
RoutingSlot{k}.kParamRoutingDest	—	9000000+	route.*	Master/Direct/Filter/None	PARTIAL
Macro0..7	macro1..4	7000000-7007000	macroN	linear	LIKELY

AssetResolver.

1. Use embedded blobs first.
2. Then search user-configured Serum folders ((Documents\Xfer\Serum 2 Presets\...), (Documents\Xfer\Serum Presets\Tables\...)).
3. Otherwise, show a "locate content" dialog.

Never redistribute resolved assets.

5.7 Measurement harness ((tools/measure_host))

1. Load (Serum2.vst3) via (juce::VST3PluginFormat); enumerate parameters (ID, name, default, steps).
2. Sweep 1025 normalized values per parameter; write (getText(v)) to (measure/serum2_params.csv).
3. Set distinct values, save state, unwrap XferJson, and diff the CBOR → (measure/key_map.json) ((module.kParam) ↔ VST3 ID).
4. Render probes (48 kHz, 2 s) and compare with Tincture: spectral distance, -3 dB cutoff, envelope crossings, LFO period.
5. Generate (serum_tables.hpp) from the outputs.

5.8 Testing strategy

- Catch2 unit tests: container parsing (fuzzed with libFuzzer), CBOR mapping, (c1m) round trip, every warp formula, filter responses, envelope timings, LFO phase, SFZ parsing.

- Golden renders at 44.1/48/96 kHz, with -100 dB RMS tolerance for deterministic paths.
- Aliasing sweep 20 Hz-20 kHz, staying below -90 dB.
- Round trips: own format must be bit-identical; Serum import → own export → import must give an identical `PatchModel`.
- pluginval strictness 10, FL Studio smoke tests, and Google Benchmark CPU figures per oscillator mode, filter family and effect.

5.9 Recommended Claude Code prompt order

1. "Read SPEC.md. Scaffold repo and CMake per 5.1/5.2; build an empty VST3 synth that passes pluginval."
2. "Implement WavetableWav (clm) + WavetableBank mipmaps + tests."
3. "Implement WavetableOsc (unison, smooth interpolation), Svf, Envelope, VoiceManager. Phase 1 acceptance."
4. "Implement all warps with dual-warp and oversampling; tests."
5. "Implement envs, LFOs, chaos, macros, 64-slot matrix, EngineSnapshot threading."
6. "Implement FilterBank families; frequency-response tests."
7. "Implement FX rack, effects, splitters, buses."
8. "Implement SerumPresetReader + Serum2Mapper (preserve unknowns); then FxpReader + migrations."
9. "Build tools/measure_host; generate serum_tables.hpp; iterate to acceptance."
10. "Implement Sample, SFZ, Granular, Spectral oscillators; Arp; Clip."
11. "Original UI, preset browser, undo/redo; final pluginval + FL Studio pass."

5.10 Open questions (measure against licensed Serum 2)

1. The full ordered filter-type list (about 107?) and each Var meaning.
2. The full Serum 2 warp enum and the math for new FM, PD and distortion warps.
3. Curves for env times, LFO rate, porta, detune per tuning mode, level dB laws.
4. `kParam*` key names per module and their VST3 IDs.
5. CBOR keys for embedded wavetables, samples, SFZ and IRs.
6. Reverb Vintage/Nitrous/Basin topologies; the distortion mode list.
7. Granular parameters and windows; Spectral FFT size, hop and timbre method.
8. The Serum 1 embedded wavetable header layout.
9. Uni Span / Uni Rand Start semantics; Ratio/Hz offset ranges.
10. Quality-mode oversampling factors.

Recommendations

- **Clear the mapping risk first.** Build the measurement harness (§5.7) in week one, in parallel with Phase 1. Mapping accuracy decides whether the hard requirement is met.
- **Make import one-way and lossless.** Import into your own documented format and keep unknown keys. Do not market a Serum preset writer.
- **Use Serum 2's own conversion as the Serum 1 path.** Map Serum 1 onto the Serum 2 model, mirroring `s1state_load`, so the engine has one compatibility target. `github`
- **Spend fidelity where users listen:** the wavetable path, unison, MG/German/French LP filters, Distortion, OTT-style multiband and Hyper/Dimension. Spectral and granular can stay approximate.
- **Stay off GPL code unless you choose GPL.** A GPLv3 release is legitimate and would let you study Vital's code directly. `github`

Caveats

- **Most compatibility facts are third-party reverse engineering, not Xfer documentation.** This covers the fxp offsets, CBOR module names, `s1state_load` and version thresholds.
 - pycabbage's results are the strongest: 25 files and byte-identical conversions.
 - The "0x3460 parameter array" claim comes from low-reputation repos and must be verified.
- **Some numbers are labelled inferences:** ~107 filter types, the cutoff range, 64 mod slots, plain-value semantics. All warp and filter equations are reconstructions, not Xfer's code.
- **I could not read the official Serum 2 web-manual pages listing filters and warps.** Feature counts come from Xfer's "What's New" PDF; some details come from reviews, forums and changelog mirrors.
- **Serum 2 is still being updated** (2.1.x states carry `version` 10.0), `pypi` so version-gate your mapper.
- **Verify JUCE and VST3 SDK licence terms at build time.** This is not legal advice; consult an IP attorney before commercial release.